

These resonating structures are called canonical forms and the actual structure lies between these three resonating structures, and is called a resonance hybrid. The resonance hybrid is represented as below.



Similar to the other electron displacement effect, mesomeric effect is also classified into positive mesomeric effect (+M or +R) and negative mesomeric effect (-M of -R) based on the nature of the functional group present adjacent to the multiple bond.

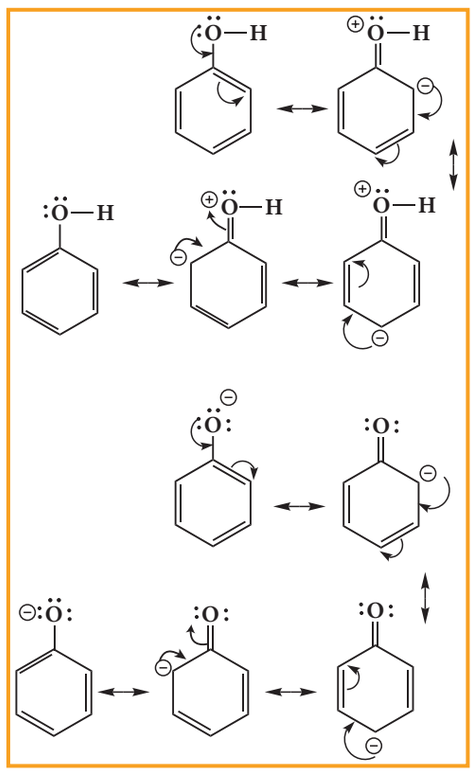
Positive Mesomeric Effect:

Positive resonance effect occurs, when the electrons move away from substituent attached to the conjugated system. It occurs, if the electron releasing substituents are attached to the conjugated system. In such cases, the attached group has a tendency to release electrons through resonance. These electron releasing groups are usually denoted as +R or +M groups. **Examples : -OH, -SH, -OR,-SR, -NH_2, -O- etc.**

Negative Mesomeric Effect

Negative resonance effect occurs, when the electrons move towards the substituent attached to the conjugated system. It occurs if the electron withdrawing substituents are attached to the conjugated system. In such cases, the attached group has a tendency to withdraw electrons through resonance. These electron withdrawing groups are usually denoted as -R or -M groups. **Examples : NO_2, >C=O, -COOH,-C≡N etc**

Resonance is useful in explaining certain properties such as acidity of phenol. The phenoxide ion is more stabilised than phenol by resonance effect(+M effect) and hence resonance favours ionisation of phenol to form H^+ and shows acidity.



The above structures shows that there is a charge separation in the resonance structure of phenol which needs energy, where as there is no such hybrid structures in the case of phenoxide ion. This increased stability accounts for the acidic character of phenol.

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